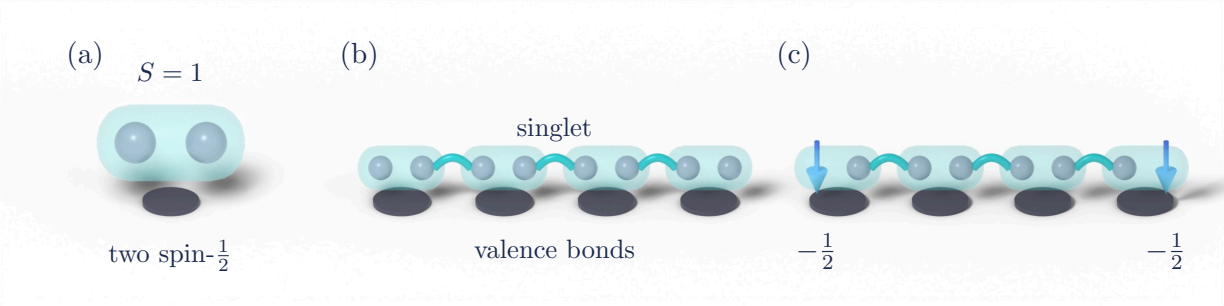

(a)
$S = 1$
two spin-$\frac{1}{2}$
(b)
singlet
valence bonds
(c)
$-\frac{1}{2}$
$-\frac{1}{2}$